Hamstring Injuries: A Paradigm for Return to Play

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Abstract

Muscle injuries are the most common sports-related injuries, with hamstring involvement most common in professional athletes. These injuries can lead to significant time lost from play and have a high risk of reinjury. We review the anatomy, mechanisms of injury, diagnostic imaging modalities, and treatment techniques for hamstring injuries. We also present the latest evidence related to return to play (RTP) after hamstring injuries, including a review of articles targeted to RTP in European soccer (Union of European Football Associations), American football (National Football League), and other professional sports. Review of imaging findings in hamstring injury, grading systems for injuries, considerations for RTP, as well as advances in injury prevention, are discussed.

Keywords
- return to play
- hamstring strain
- injuries
- soccer
- football

Muscle injuries are the most common injuries associated with playing sports, particularly those engaged in sports requiring high-speed activities and dynamic movements. These injuries can lead to long-term impairment and have a high risk of reinjury.[1] Injuries to the hamstring muscles are the most common injury in soccer, the world's most popular sport, and they are often encountered by physicians diagnosing and treating athletes at any level.[2] These injuries are also common in American football, rugby, baseball, field hockey, dancing, and skiing.[3] The average practice and game time loss due to hamstring injury is between 17 and 60 days, contingent on the grade of injury.[4]

In addition to their frequent occurrence, high recurrence rates further complicate the management of these injuries and add to adverse outcomes in terms of return to play (RTP) and athletic performance.[5] At the professional level, this can result in significant financial consequences. In 2021, hamstring injuries were the most frequent and expensive injury across the five major European soccer leagues, accounting for nearly $100 million in lost salary.[6] Eliakim et al estimated that an average English Premier League team loses ~£45 million per season due to injury-related decrements in performance.[7] The diagnosis and treatment of these injuries are complex, as are the management options following injury.

Of particular importance is the decision-making process for athletes returning to play after a hamstring strain. This decision is highlighted by the need to balance risk of reinjury and long-term damage with the coach's and athlete's desire to resume competition. The result is a very difficult situation because often competing interests between involved parties further complicate the factors that must be considered.

From the athlete's side, the desire to compete, to help the team on the field or court, is weighed against concerns for further injury that at the professional level can have significant monetary implications in terms of new contracts or performance-based bonuses. From the team perspective, many factors, such as the athlete's role on the team, the timing of the injury during the season, contractual implications, and varying coaching, front office, and ownership opinions all contribute to the decision about when an athlete should return to competition. From the physician's perspective, balancing the healing process with the risk of reinjury are of central importance in advising athletes and teams on the best time for an athlete to resume participation.

All these considerations are further complicated by the differences in sport and in competition level, including return to partial participation and activity level versus unrestricted return to game competition. Additional outside

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voices in professional sports are also factors, ranging from the media, fans, and athlete's personal relationships, which all contribute to the environment in which these decisions are made, with far-reaching economic implications.

This article describes the current body of knowledge surrounding the radiologic assessment of hamstring injuries in athletes, with a particular focus on professional athletes. Combined with appropriate management decisions, this information determines the implications for RTP decisions.

Anatomy and Pathology

The posterior compartment of the thigh includes the semimembranosus, semitendinosus, and biceps femoris muscles, referred to as a group as the hamstring muscles (►**Fig. 1**). These muscles play a crucial role in the dynamic stability and functional movement of the lower extremity because they span two joints: the hip joint and knee joint. The hamstring muscles converge to robust tendons with an origin at the posterior medial aspect of the ischial tuberosity and insertion on the femur and tibia distally. The semitendinosus and long head of biceps femoris share a common origin, referred to as the conjoined tendon, with the semimembranosus origin separate and anterolateral from the conjoined tendon attachment.[8] Of note, authors vary in the precise description of the conjoined tendon attachment.

Biceps Femoris Muscle

The biceps femoris muscle is divided into two heads: the long head arising from the medial facet of the ischial tuberosity and the short head arising from the lateral proximal femur at the linea aspera.[9] The tendons of the short head and long head merge above the knee joint, and the combined tendons insert into the head of the fibula, crural fascia, and proximal tibia.[10] Injuries to the long head of the biceps femoris are the most common hamstring injury and were described to account for up to 80% of all hamstring strains.[11,12]

Semitendinosus Muscle

The semitendinosus muscle lies in the posteromedial thigh and runs from a proximal origin on the ischial tuberosity to its distal insertion as part of the pes anserinus.[8] It creates a conjoined tendon with the long head of the biceps femoris, forming an aponeurosis.

Semimembranosus Muscle

The semimembranosus muscle lies posteromedial in the thigh adjacent to the semitendinosus, with its origin at the anterolateral ischial tuberosity extending distally to its insertion as part of the pes anserinus where it descends under the semitendinosus.

Hamstring Injury and Classification

Hamstring injuries can occur at the proximal or distal tendons, the myotendinous junctions, or at the central intramuscular tendon.[13]

Proximal hamstring injuries can be divided into acute relatively high-energy tensile overload injuries, more common with sprinting, and lower energy tensile overload injuries, commonly seen with dancers while stretching.[14,15] Hamstring injuries most commonly occur while sprinting in full stride or attempting to overstride. The most common site of injury is the proximal myotendinous junction of the long head of the biceps femoris (►**Fig. 2**). Lower energy tensile overload injuries commonly involve the proximal myotendinous junction of the semimembranosus.

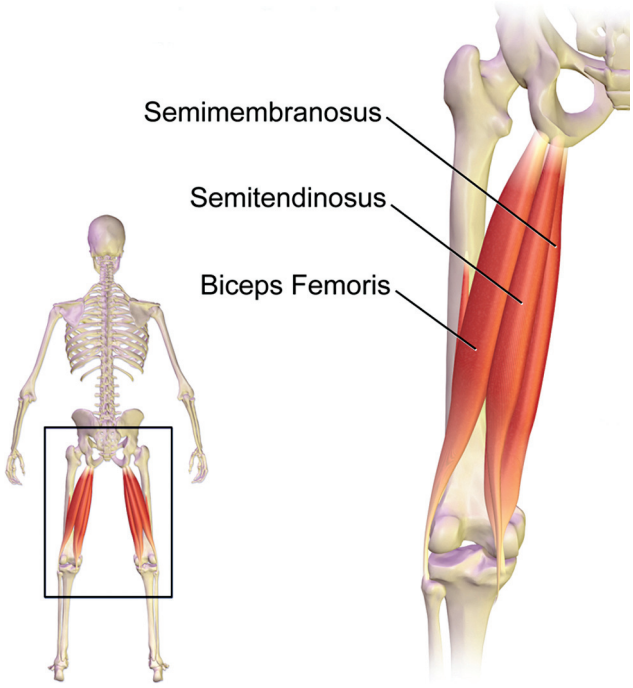

Fig. 1 Basic hamstring anatomy.

Fig. 2 A 25-year-old professional football player with an acute right hamstring injury during a game. (**a**) Coronal short tau inversion recovery image of a grade 1 to 2 strain of the right biceps femoris centered at its proximal myotendinous junction with ~ 25% fiber disruption on the short axis and edema spanning 23 cm in the craniocaudal dimension. (**b**) Four weeks later, coronal short tau inversion recovery image of a recurrent right hamstring strain after an acute injury during practice.

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Various classification systems grade the severity of muscle injury, with particular attention given to hamstring injury classification.[16] Grading systems aim to categorize and describe muscle injuries, yet they currently lack uniformity. Primary types of grading systems are classified into three types:

1. Clinical grading systems: Based on indirect evaluation of muscle, including signs and symptoms such as pain, swelling, ecchymosis, and whether a palpable defect is present.
2. Systems based on magnetic resonance imaging (MRI): Evaluation on the extent of muscle involvement visible on MRI, as well as location, as major determinants of injury severity.
3. Ultrasonographic (US) grading systems: Imaging techniques to visualize and grade the injury that have evolved over time with technological improvements and clinical understanding of injuries.

In 2014, Pollock et al proposed the British Athletic Muscle Injury Classification (BAMIC).[17] This system was designed to expand on the previously most accepted muscle injury classification that described injuries as minor, moderate, or complete. It used MRI to classify muscle injury into grades 0 to 4. Grade 0 represented muscle injuries that were not visible on imaging, although these injuries were believed to be structural, yet below the resolution of imaging. Grades 1 to 4 were thought to represent muscle injury or tear, recommending against the use of the word "strain."

These injuries were subdivided by longitudinal length and by the percentage of cross-sectional area of the high T2 signal on MRI. Grade 1 injuries involve < 10% of the muscle with a longitudinal length < 5 cm. Grade 2 injuries involve 10 to 50% of the muscle and extend between 5 and 15 cm. Grade 3 injuries are > 50% of the muscle cross-sectional area or > 15 cm in length. Grade 4 injuries are complete tears of the muscle or tendon, often with a palpable gap.

Further subclassification is by location: "a": myofascial injury; "b": injury within the muscle belly; and "c": intratendon involvement. An additional descriptor denotes proximal, central, or distal third involvement. Specific location criteria for hamstring injuries was discussed, with proximal third injuries located above the lower margin of the gluteus maximus and distal third being below the origin of the short head of biceps femoris.

Many other classification systems have been developed to classify muscle injuries.[18] Shelly et al divided muscle strains on MRI into grade 1 strains with muscle edema (no muscle tear), grade 2 strains with partial tears of the muscle (< 50% of the cross-sectional area [CSA]), and grade 3 strains with near complete (> 50%) or complete muscle tears.[19]

Acute proximal hamstring muscle complex injuries were classified by Carmichael et al[20] into five types of hamstring injuries. Type 1 injuries are osseous avulsions, usually apophyseal injuries in skeletally immature patients. Type 2 injuries are at the musculotendinous junction. Type 3 injuries are incomplete tendon avulsions from the bone (►**Fig. 3**). Type 4 injuries are complete tendon avulsion with no or minimal retraction of the tendon ends. Type 5 injuries are complete tendon avulsions from bone with retraction of the tendon ends. Then injuries can be further subdivided into type 5a injuries, not associated with scarring to the sciatic nerve, and type 5b injuries, associated with sciatic nerve tethering. This classification system has been described as incorporating the most detailed anatomical information and the most clinically useful for proximal hamstring injuries requiring repair.[21]

Other classification systems often incorporate a combination of these systems, including mechanism of injury, anatomical location based on MRI and US features, clinical presentation, and even recently the use of artificial intelligence (AI) to enhance prognostic value and improve diagnosis, management, and evaluation for return to competition.[22]

In 2016, an expert consensus group developed a new classification system for muscle injuries, focusing specifically on hamstring injuries.[23] This system has four main categories: mechanism of injury (M), location of injury (L), grading of severity (G), and number of muscle reinjuries

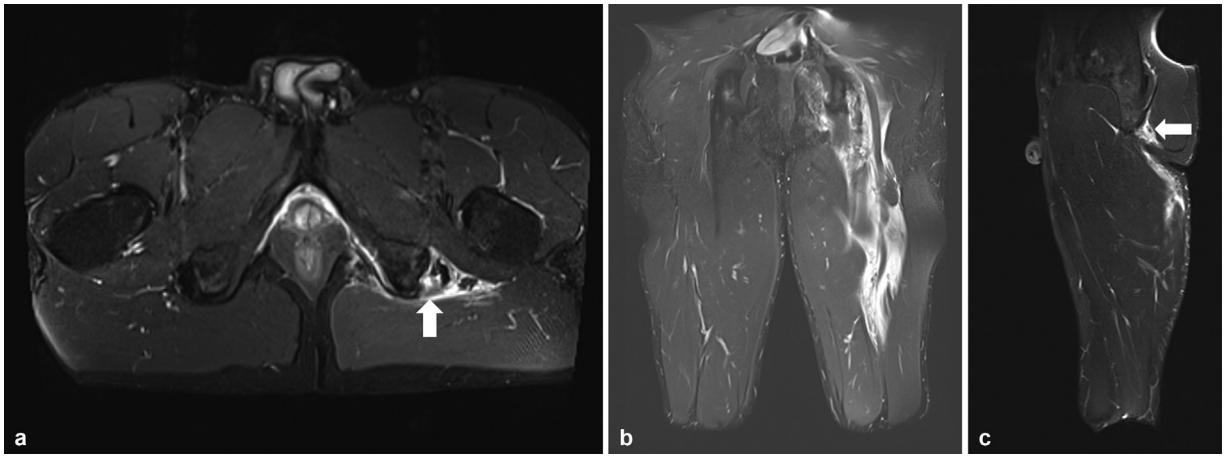

Fig. 3 A 40-year-old professional football player with acute hamstring injury. (**a**) Axial short tau inversion recovery (STIR), (**b**) coronal STIR, and (**c**) sagittal STIR sequences demonstrating a complete avulsion of the left conjoint tendon origin with ~ 2.8 cm of retraction and 50% partial tearing of the left semimembranosus origin. There are grade 1 muscle strains of the proximal semimembranosus and mid-biceps femoris muscles. White arrows (a) and (c) denote site of avulsion at ischial tuberosity.

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(R), abbreviated as MLG-R. Direct and indirect muscle injuries were considered in different ways, and the location of the injury was also incorporated, including proximity to the myotendinous junction attachment, considered to increase injury severity. Direct injuries were defined as an injury caused by an external force, whereas indirect injuries were defined as stretch-induced injuries caused by a sudden forced lengthening over the viscoelastic limits of muscles occurring during a powerful contraction.[24] The authors found that MRI parameters including the percentage of CSA, the craniocaudal length, and the injury volume all correlate to injury time loss and may help guide predictions about RTP.

Additional muscle injury classification systems were proposed by O'Donoghue (1962), Ryan (1969), Takebayashi (1995), Peetrons (2002), and Stoller (2007), and in general have three to four grades and aim to describe the extent of injury.[24] In 2012, the Munich consensus statement aimed to consolidate these and other classifications into a unified system, reflecting survey data from a group of sports medicine experts. This classification uses a combined approach of both clinical and imaging assessment and further divides injuries into types 1 to 4 based on clinical signs, location, and imaging findings by US or MRI.

Risk Factors for Hamstring Injury

The current understanding of hamstring injuries suggests that several factors contribute to the etiology of strain injury. Anatomical factors, such as the fact that the hamstrings cross two joints, allowing for simultaneous extension at the hip and flexion at the knee, place increased demands on the hamstrings that may contribute to injury. High levels of eccentric force and muscle strain are associated with hamstring injury.[25]

Research has suggested that dual innervation of the biceps femoris may predispose to asynchronous muscle contractions, with the tibial branch of the sciatic nerve innervating the long head while the common peroneal branch innervates the short head, which may contribute to an increased risk for injury.[26] Increasing age, prior injury, strength imbalances, extremes of flexibility, and muscle fatigue have also all been implicated as risk factors for hamstring injury.[27]

Imaging Modalities

Imaging is fundamental in assessing the location, extent, and severity of hamstring injuries, allowing for appropriate diagnosis and management for athletes. In the acute injury setting, the initial indication for imaging is to exclude proximal hamstring avulsion that is best managed with early surgical repair.[28,29]

Magnetic Resonance Imaging

MRI allows for detailed soft tissue evaluation of hamstring injuries. It not only helps confirm the clinical diagnosis of strain but also provides detail on the location, CSA, and extent of tearing. MRI provides a large field of view and high resolution, allowing for the identification of muscle strains, tears, and tendon avulsions, as well as evaluation of associated bone injuries. It also allows for image-based grading of the strain[19] and for classification of the injury pattern. MRI and US are equally sensitive in assessing hamstring injury, but MRI offers a more detailed analysis of the injury and is less user dependent. The higher cost of MRI is generally not a precluding factor for professional sports teams, but it may be a factor at the high school and college level.

Hamilton et al showed excellent interobserver and intraobserver reliability for grading and predicting MRI parameters in acute hamstring injuries using the Peetrons grading system.[30] This implies a high degree of consistency and reproducibility in scores across different evaluations or evaluators, supporting the use of MRI for accurate diagnosis and prognosis. Overall, MRI is considered the first-line imaging for initial injury and for follow-up examinations.

At our institution, we have developed a specialized MRI hamstring protocol, highlighted by a composite image after coronal acquisitions at two table positions, allowing for delineation of the entire hamstring complex (►**Fig. 4**). It allows for visualization of the entire extent of the injury in one sequence. This specialized MRI protocol provides a comprehensive view of the hamstring complex, enabling better visualization and assessment of the injury.

Ultrasonography

US is increasingly recognized as a dynamic and cost-effective tool for evaluating muscle injury, allowing for real-time visualization of muscles and tendons with the added capability of imaging during movement. It also offers several

Fig. 4 Magnetic resonance imaging (MRI) hamstring imaging protocol. Specialized MRI of the hamstrings. (**a**) Composed short tau inversion recovery (STIR) image from an American football player after coronal acquisitions at two table positions delineating the entire hamstring complex. There is a grade 1-2 injury centered at the proximal myotendinous junction of the right biceps femoris. (**b**) Similar imaging technique with a composed coronal STIR image from another American football player with an acute hamstring injury. There is a mild grade 1 strain at the right biceps femoris muscle belly proximally.

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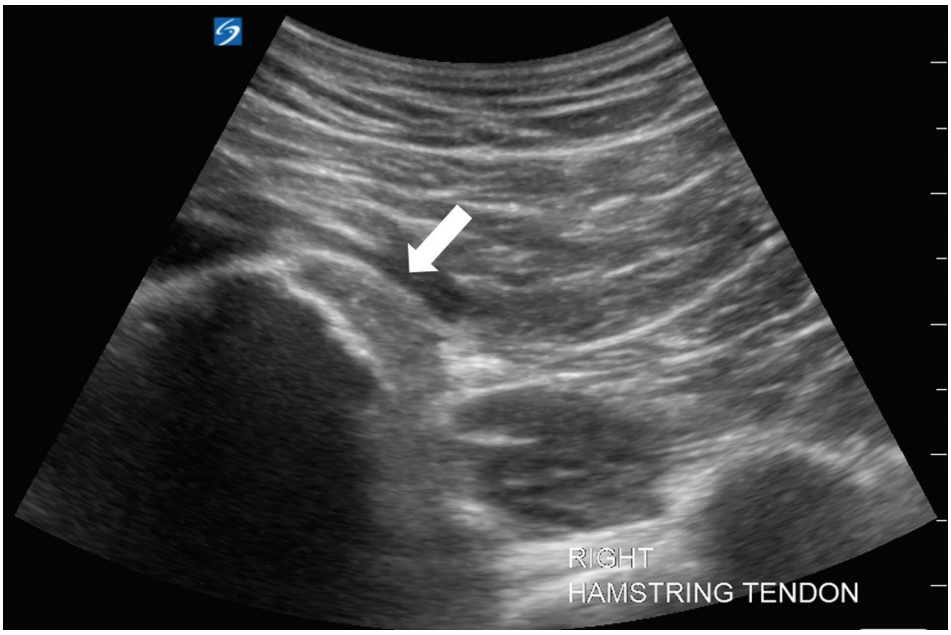

Fig. 5 Ultrasonographic images of hamstring origin. A 26-year-old professional football player with intact right hamstring origin (arrow).

advantages, such as the absence of radiation, portability, and good spatial resolution (►**Fig. 5**). It can effectively detect acute injuries including muscle tears and tendon avulsions, as well as visualize adjacent hematomas and edema. US can also be used to guide interventions, such as percutaneous aspiration and injections. Low cost, few contraindications, and higher resolution than MRI also make US a viable option for evaluating a hamstring injury.[31]

Ultrasonography versus Magnetic Resonance Imaging

Connell et al compared US and MRI for assessing hamstring injuries in Australian professional football players at 3 days, 2 weeks, and 6 weeks after injury, allowing for comparison of the same injury on both modalities. They noted that in the acute phase of the injury, both sonography and MRI showed high sensitivity for hamstring injuries.[32] For those requiring follow-up imaging, MRI was preferred for its improved contrast resolution and fine detail in assessing evolving injury patterns over time. Specifically, the ability to follow the resolution of T2 hyperintense signal and formation of low-signal scar tissue makes MRI a better option. The authors suggested that some of this subtle detail may be related to the differences in the proton and hydrogen environment caused by the formation of ionic bonds during the healing process, resulting in susceptibility artifacts that can be visualized on MRI.

The biceps femoris intramuscular tendon was found to be the most common site of injury, and sonography was not as sensitive as MRI for localizing tears along the central intramuscular tendon. It instead localized these tears as being in the muscle belly, which may have led to only moderate agreement between MRI and US for identifying the site of injury.

On US, the CSA of injury was a useful predictor for RTP. Injuries appeared significantly larger on MRI than on sonography at all time points due to its ability to show subtle edema. The longitudinal length of hamstring tear on MRI had the highest statistical correlation with recovery and was the best radiologic predictor for return to competition. The authors noted that a player's RTP ability depends on many factors in addition to what is seen on imaging, including the ability to sprint pain free, the absence of pain and stiffness during a training session, as well as factors intrinsic to that player's health and additional team considerations.

Analysis of MRI showed longer time to recovery in injuries involving the biceps femoris muscle, increased length of the injury, increased cross-sectional injury area, and injury outside of the musculotendinous junction. With US, the presence of a hematoma was an additional contributing factor to increased length of recovery. Despite this, a high percentage of players with residual abnormal imaging findings at 6 weeks were symptom free and had returned to competition.

Normalization of Magnetic Resonance Imaging Findings

The average time missed due to hamstring injury depends on the grade of the injury, with an average of 17 days missed for grade 1 injuries, 22 days with grade 2 injuries, and 73 days for grade 3 injuries in European professional soccer players.[33] The median duration for RTP in Australian football players was 21 days (range: 4–56 days).[32] However, athletes often become symptom free within 10 days after an injury, even though the histopathologic healing process takes weeks to months.[34]

In the previously described study by Connell et al, some players showed larger injuries on the 2-week follow-up MRI

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and US. At 6 weeks, 35% of players still had residual findings on MRI, although all but one were asymptomatic and had returned to play. A study of Dutch athletes, including professional football players, found that 89% of clinically recovered hamstring injuries showed intramuscular increased signal on MRI. In addition, 42% demonstrated abnormal intramuscular low signal, compatible with new fibrous tissue.[35]

Despite athletes often feeling symptom free within 10 days postinjury, imaging findings take longer to normalize, likely reflecting the healing and repair process. This lag in MRI normalization suggests the imaging findings correlate with the healing process of the muscle/tendon and may in part explain the high recurrence rates in these injuries.

Hamstring Injury Management Options

Conservative Management

Conservative management of hamstring injuries involves rest, pain management, and rehabilitation. This approach is commonly used for mild to moderate hamstring injuries (grade 1 and some grade 2 strains). Therapeutic modalities like ice and compression can be used to handle initial inflammation, pain, and swelling after an injury. The classic acronym RICE (rest, ice, compression, and elevation) is adopted in this setting. In addition, the use of nonsteroidal anti-inflammatory drugs can help in the short term with inflammation and pain.

Physical therapy is an important component of rehabilitation for hamstring strain injuries. Physical therapy integrates stretching, strengthening, stabilization, and functional exercises. Guidelines suggest using progressive agility and trunk stabilization exercises that can significantly reduce the reinjury rate following a hamstring injury.[12] Hamstring-strengthening exercises, including eccentric exercises, should be initiated early in the rehabilitation process and be guided by the patient's pain tolerance. There is also support for progressive running programs that involve acceleration and deceleration.

Athletes treated at our institution are often part of the same standardized six-stage rehabilitation program. Stage I of the protocol focuses on functional movement progressions and range of motion exercises. Stages II and III focus on core stability exercises, agility, and the gradual resumption of high running velocities. Later stages involve sports-specific sessions and functional testing on the field or court. To pass final clearance criteria, the athlete is required to perform unrestricted sports-specific functions without any limitations or symptoms. These final sports-specific functions depend on the type of sport and are tailored toward basketball, baseball, football, lacrosse, soccer, and running/track.

Increasingly essential in the later stages of rehabilitation, using sport-specific movements to build toward full participation has become an important component of the rehabilitation process. These movements aim to improve proficiency in the unique movement patterns and demands of a particular sport, moving beyond general physical therapy exercises.

Platelet-Rich Plasma and Hematoma Aspiration

A study from our group focused on the treatment of grade 2 hamstring strains among athletes and its influence on their RTP timeline.[5] Most grade 2 injuries are associated with at least a small hematoma that can result in pain related to mass effect on the sciatic nerve. This has been referred to as "hamstring syndrome" and can result in delayed recovery, decreases in function and strength, as well as chronic pain due to formation of adhesions and entrapment of the nerve by scarring.[36] Combining hematoma aspiration with platelet-rich plasma (PRP) injection treatments was compared with the outcomes of the conventional treatment of rehabilitation alone (►**Fig. 6**). Athletes who underwent the combination of hematoma aspiration and PRP injection had quicker RTP and a lower rate of recurrence compared with those who were treated with conventional rehabilitation, with a mean time to RTP of 42.5 days in the control group and 26.7 days in the PRP group. This suggests that the consideration of such combined treatment in selected cases may yield better outcomes in terms of RTP timelines and recurrence rate.

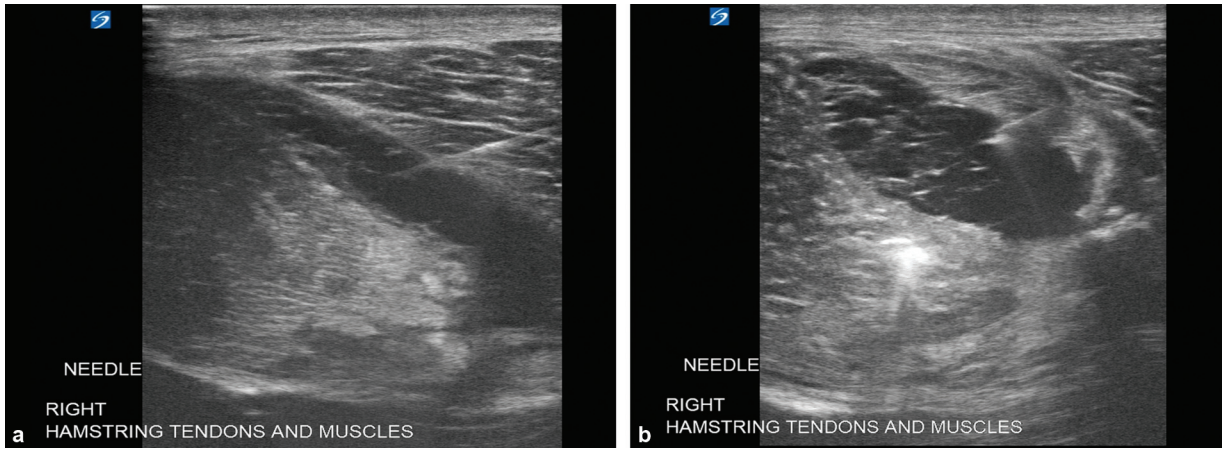

Fig. 6 A 25-year-old professional football player with a moderate intra- and intermuscular hematoma following a high-grade tear (grade 2 to 3 strain) of the right proximal hamstring centered at the myotendinous junction of the conjoined tendon. (**a, b**) Ultrasonographic images showing needle aspiration and injection of the right hamstring hematoma (fluid collection: 12 mL).

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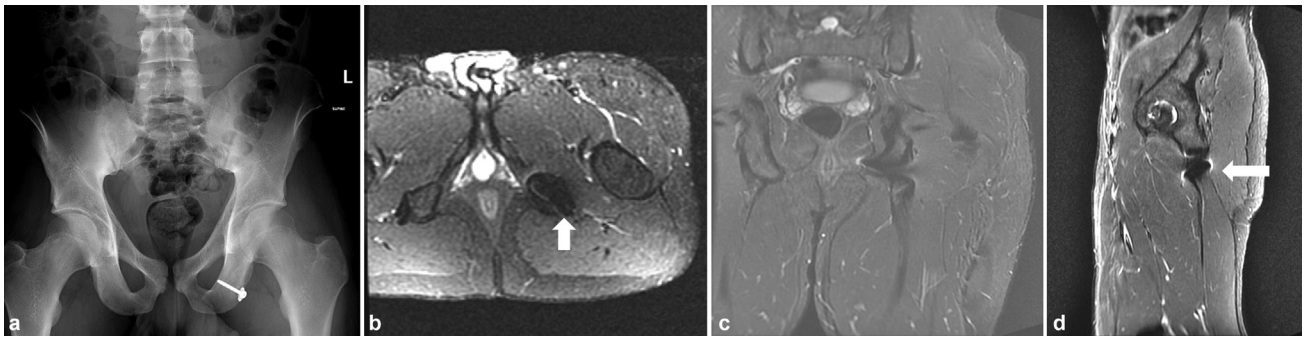

Fig. 7 A 22-year-old professional baseball player with a history of left ischial tuberosity avulsion fracture, treated surgically. (**a**) Frontal radiograph of the pelvis demonstrates intact postsurgical changes with a single screw paced in the left ischial tuberosity. (**b**) Axial short tau inversion recovery (STIR), (**c**) coronal STIR, and (**d**) sagittal STIR of the left hip demonstrate intact single screw in the left ischial tuberosity with mild tendinosis of the hamstring tendon origin without a tear. White arrows (b) and (d) denote ischial tuberosity screw with susceptibility artifact.

Professional and college team physicians are increasingly using PRP injections, with survey data demonstrating 93% of team physicians using PRP in their practices, with hamstring injuries the most commonly treated condition.[37]

Surgical Management

Surgery is indicated for complete tears or refractory injuries, especially in the setting of avulsion injuries at the hamstring tendon origin (►**Fig. 7**). Surgical techniques include open repair, endoscopic repair, and augmentation procedures. Preoperative imaging, often involving MRI and US, aids in delineating the injury's characteristics and assists the surgical team in planning the procedure. Postoperatively, imaging plays a vital role in monitoring for complications and guiding rehabilitation.[1] Small series have described surgical intervention for professional athletes who failed at least 3 months of conservative treatment, with favorable outcomes in a series of 10 patients.[13]

Return to Play and the Union of European Football Association Study

The most comprehensive study in professional athletes following hamstring injuries involves the Elite Club injury study by the Union of European Football Associations (UEFA). It began in 2001, with the initial study analyzing the patterns and trends of hamstring injuries in European soccer over a 13-year period.[38] Of the reported injuries, 65% occurred during matches; 35% happened during training sessions, suggesting increased risk during matches, likely due to the increased intensity. Most of the hamstring injuries (54%) were of moderate severity, resulting in an average of 17 days of missed time. The study also found limited implementation of injury prevention measures among the clubs.

The 2016 publication of the study analyzed retrospective data from MRI documentation of 255 hamstring injuries, specifically focusing on grade 1 and 2 injuries.[39] The authors concluded that the radiologic grading was significantly related to the time to RTP, with grade 2 injuries requiring an average of 6 more days to RTP than grade 1 injuries. Weak correlations were found between the size of the edema on MRI and the RTP for both grades of injuries. The study suggested that radiologic grading of hamstring injuries can be helpful in prognosticating RTP in soccer.

The 2022 publication described a continued increase in incidence of hamstring injuries during training and matches, rising from 12% of injuries in 2001–2022 to 24% in 2021–2022, with 18% of hamstring injuries representing recurrent injury. Additionally, the authors focused on the advances in imaging technologies since the early days of the study and how they have influenced the diagnosis of hamstring injuries.[40] The data were again drawn primarily from players on teams that qualified for the group stage of the UEFA Champions League in different seasons. Comparing the time periods of 2007–2011 with 2011–2018, a contrasting pattern emerged, with more recent years (2011–2018) showing an increased rate of structural injuries of 72% (defined by evidence of muscle tears evident on MRIs) versus 30% for the 2007–2011 time period.

A key difference over these periods is attributed to the evolution in the quality of MRI scanners. This technological advancement has not only improved image quality but also shifted how injuries appear on MRIs. Injuries are more likely to show as grade 2 (structural) than grade 1 (functional). This trend highlights the ability of MRI to continue to play an increasing role in the evaluation of injuries. An additional finding of a high rate of recurrent hamstring injury within 2 months of RTP suggests the need for extreme caution during this stage of recovery.

The UEFA club study is to date the most comprehensive study on hamstring injuries in professional athletes. It provides valuable insights into the incidence, burden, and trends of hamstring injuries over a 21-season period. Although offering insight into the trends of the injury rates and the overall burden of hamstring injuries, the study is limited in its ability to explain the underlying factors contributing to these trends.

Comparisons of the BAMIC with specific hamstring injuries have been the subject of study in recent years. English Premier League club players were evaluated for return to full training after a specific anatomical injury location was assessed, showing that hamstring muscle injuries involving

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the intramuscular tendon (BAMIC "c" category) were found to have longer return to full training time (36 days versus 24 days without tendon involvement) and a significantly increased reinjury rate (38% reinjury rate versus 12.5% without tendon involvement).[41] Additional studies found correlations between higher BAMIC grade hamstring muscle injury and increased RTP time in professional soccer and elite track and field athletes.[42,43] Using the BAMIC system, rehabilitation protocols have been developed that provide guidance for management of these injuries, assisting medical staff decision making.[44]

Return to Play in American Football: The National Football League

The National Football League (NFL) is the most popular sports league in the United States, averaging > 17 million viewers during the regular season.[45] NFL players have a 41% chance of sustaining a lower extremity injury during the course of a season, with hamstring injuries the most common.[46]

MRI has been used to evaluate RTP in NFL players. Defensive backs, linebackers, and wide receivers are at particularly high risk for these injuries, likely due to the nature of the positions, with high rates of sprinting. Authors at our institution aimed at correlating MRI findings with RTP time in professional football players after acute hamstring strains and proposed a scoring system to predict anticipated time missed.[18] On average, players missed 2.6 games after acute hamstring strain injuries.

The MRI score, ranging from 2 to 19 (higher score correlating with increased injury severity), factored in the patient's age at the time of injury, the number of muscles involved, location of the injury, presence of insertional injury, cross-sectional percentage of muscle injured, presence of retraction of muscle or tendon, and the length of the long-axis T2 signal (►**Table 1**).

Additionally, presence of a fluid collection or hematoma, any signs of chronic tendinopathy, peritendinous or perimuscular edema, or intramuscular cysts were documented. An increased MRI score was found to correlate with a higher number of games missed, with other significant predictors including percentage of muscle/tendon involvement, number of muscles involved, and the amount of retraction. Traditional MRI grading was also performed that showed a significant difference in games missed between grades 1 plus 2 injuries compared with grade 3 injuries, but no difference between grades 1 and 2.

Australian Rules Football

Slavotinek et al explored the prognostic value of MRI in Australian rules rugby athletes, evaluating 37 athletes suspected of having hamstring injuries who underwent clinical assessments and MRI.[47] They found an association between the rehabilitation time and MRI parameters of muscle injury, including the approximate volume of muscle injury and the extent of cross-sectional muscle involvement. Specifically, cases with abnormal muscle CSA > 50% were associated with longer recovery times. The data suggest that the rehabilitation time is related to the proportion of muscle units disrupted during injury. Of note, the specific muscle involved did not significantly impact the rehabilitation time for this sport. This finding suggests no individual muscle has greater functional significance or different healing time.

Clinical Parameters Used for Return to Play

Numerous studies have evaluated the use of clinical information for making RTP decisions. These studies have explored various factors such as injury severity, location, and presence of symptoms to determine the prognosis and potential risk of reinjury.

Erikson and Sherry developed a clinical algorithm to determine RTP parameters, including tenderness to palpation, strength testing with an accepted difference between legs of < 5%, active hamstring test (H-test), single-leg bridge test, and sport-specific movement testing.[48] The active hamstring test is performed by an athlete doing a straight leg raise as fast as possible. The single-leg bridge test is performed by an athlete lifting hips off the ground while keeping one leg extended. Athletes were cleared when able to complete these tests without pain or hesitation.

De Vos et al found that the number of previous hamstring injuries, active knee extension deficit, isometric knee flexion force deficit at 15 degrees, and presence of localized discomfort on palpation after RTP all correlate with higher rates of reinjury.[49] These studies suggest that a comprehensive assessment of clinical information is crucial in making RTP decisions after hamstring injuries. Incorporating objective measures such as strength testing and sport-specific

Table 1 Hamstring magnetic resonance imaging scoring system

Points	Age, y	Muscles Involved, n	Location	Insertion	Muscle Injury, %	Retraction, cm	Long-axis T2 signal length, cm
0				No	0	None	0
1	≤ 25	1	Proximal		25	< 2	1–5
2	26–31	2	Middle	Yes	50	≥ 2	6–10
3	≥ 32	3	Distal		≥ 75		> 10

Note: Adapted with permission from Cohen SB, Towers JD, Zoga A, et al. Hamstring injuries in professional football players: magnetic resonance imaging correlation with return to play. Sports Health 2011;3(5):423–430.

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movement testing can provide insight into an athlete's readiness to return to sport.

Injury Prevention

The UEFA Elite Club study also evaluated injury rates among football clubs participating in the UEFA Champions League, gaining insights via a survey reviewed by a medical team from each club.[50] They primarily evaluated the implementation and effectiveness of the Nordic Hamstring Exercise (NHE) program in these clubs during the 2020–2021 season, comparing it with 2012–2014 data.

The NHE program is an eccentric exercise performed while on the knees with the ankles held. The athlete lowers their upper body toward a prone position as slowly as possible. This exercise is generally performed with one motion, as the athlete catches themselves on the floor with their hands. However, newer adaptations of the exercise have been performed as eccentric-concentric exercises, with the athlete performing a full NHE to the floor and back without a break. Some studies show the NHE can decrease the risk of hamstring injuries by 50%.[51,52] The UEFA study demonstrated that the teams using the NHE program for all or most of their players had fewer hamstring injuries and a lower injury burden versus the teams that only used the program for individual players with current or prior hamstring injury.

Overall, adoption of the NHE program is low among professional soccer clubs and also true of American professional sports teams, likely due to several reasons. Team physicians, training staff, and coaches may not unaware of the data showing decreased injury rates and injury burden in athletes using the program. Most of the research on the NHE program has been performed in soccer athletes, which may lead to lower adoption in other sports that have not seen correlation with decreased injury burden in their own sport. Additionally, even if teams are aware of the benefits, it can often be challenging to implement injury prevention techniques into a busy training and practice schedule. Athletes also report soreness following these exercises, which can further decrease the desire to incorporate the program consistently.[53,54]

Another factor shown to play a role in hamstring strain injury is hamstring strength. Opar et al demonstrated that risk of hamstring strain injury decreases by up to 8.9% for every 10 N increase in eccentric force strength.[27] Van Dyk et al found that eccentric hamstring strength imbalances between legs were associated with a higher risk of reinjury after RTP following a hamstring injury.[55] They showed that athletes with a greater imbalance in eccentric hamstring strength between legs had a significantly higher risk of reinjury compared with those with more balanced strength ratios, which were evaluated by calculating the peak concentric and eccentric hamstring strength of the injured leg and comparing it with the peak strength of the uninjured leg.

Muscle architecture contributes to the force production capability of the muscle, as well as the velocity at which the muscle can contract. Muscle architecture has been described as the fascicle length, pennation angle, and muscle thickness or anatomical CSA. Hamstring strain injuries have also been linked to poor neuromuscular control and coordination. Neuromuscular control and coordination deficits can lead to improper movement patterns and increased strain on the hamstring muscles during dynamic activities.

Artificial Intelligence and Machine Learning

Recent advances in AI have opened doors to new possibilities in predicting the outcome and prognosis of hamstring muscle injuries, particularly in professional athletes. These technologies allow for the analysis of large data sets, including athlete demographics, injury characteristics, and imaging findings, allowing for identification of potential risk factors and developing predictive models for RTP. Several studies have explored using AI to help determine RTP implications for hamstring injuries.

Ayala et al developed a machine learning model that incorporates preseason psychological and neuromuscular measures to predict hamstring injuries in professional soccer players.[56] The study found that the model had moderate to high accuracy in identifying players at risk of hamstring injuries during preseason screenings. Similarly, Mandorino et al adopted machine learning techniques to predict noncontact injuries in professional soccer players using a combination of physical and performance data.[57] Their model achieved an accuracy of 87% in predicting noncontact injuries, including hamstring injuries. Valle et al developed a RTP prediction system using the MLG-R injury classification system on Football Club Barcelona soccer players, using three different statistical and machine learning approaches that showed excellent predictive power for mean RTP time.[58] The use of AI in sports medicine and predicting RTP is discussed in more detail in a separate article in this issue.

Conclusion

Hamstring injuries are a common occurrence in professional athletes, particularly in sports that involve sprinting, jumping, and rapid changes in direction. These injuries can range from mild strains to complete tears, and they often require a significant amount of time for recovery and rehabilitation. Current considerations for RTP after hamstring injuries focus on several key factors, such as the severity of the injury, individual healing rates, and the athlete's ability to meet specific criteria before being cleared for full participation in their sport.

Numerous studies have examined the optimal management and RTP protocols for hamstring injuries in professional athletes. These studies have highlighted the importance of using a multidisciplinary approach that includes close collaboration between medical professionals, athletic trainers, coaches, and the athletes themselves.

In particular, MRI has emerged as a valuable tool in the evaluation and management of hamstring injuries. MRI allows for accurate assessment of the extent of the injury, identification of specific muscle involvement, and monitoring of the

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healing process over time. Moreover, MRI can provide valuable information on potential risk factors for reinjury and guide the development of individualized rehabilitation programs. MRI scoring systems are currently used to assess the severity of hamstring injuries and to predict the time required for RTP. US provides another tool that can evaluate for injury in the acute setting, and it can also be used for image-guided intervention that has proven to decrease the recovery time after an injury.

The UEFA hamstring study, a longitudinal study over 20 years, has provided key insight into the increasing prevalence of these injuries, the impact on player performance, and the need for standardized RTP protocols. Studies evaluating RTP in the NFL have also shed light on the challenges faced by teams and medical staffs in determining the optimal timing for RTP in professional football players. Australian rugby players, who often experience high rates of hamstring injuries, have also been the subject of research examining RTP protocols and the role of MRI in guiding these decisions.

New advances in AI are likely to play an increasing role in the future of RTP protocols for hamstring injuries. These technologies have the potential to analyze large data sets and generate predictive models that can assist health care professionals in making more accurate and individualized decisions regarding RTP.

Overall, the literature highlights the need for a multidisciplinary approach to determine the optimal timing and criteria for RTP after hamstring injuries in professional athletes, considering clinical evaluation, imaging findings, individual factors (such as previous injury history and physical readiness), and collaboration between medical professionals, coaches, and the athletes themselves.

Conflict of Interest
None declared.

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